

01 CASE OBJECTIVE

Creative software environments involve complex workflows where users may perform multiple actions before completing an intended task. From a **Quality Engineering** perspective, quality should not be evaluated only through software functionality or visual appearance. It should also consider whether users can complete creative workflows efficiently, reliably and consistently while avoiding unnecessary errors and rework.

This case study proposes a **Quality Engineering** approach that combines user behaviour, workflow data and direct feedback to identify quality concerns within creative workflows. The improvement process then moves through structured analysis, experimentation, risk evaluation, controlled validation and continuous improvement.

02 QUALITY PROBLEM

A creative workflow can lose quality when users repeatedly encounter delays, errors, unnecessary steps or rework during task completion. Such problems may not always originate from a single visible feature. They can emerge from interactions between workflow stages, user behaviour, system conditions and the complexity of the task itself.

The proposed approach therefore focuses on understanding where quality problems occur, why they occur and how strongly they affect workflow performance. Instead of improving a feature based only on isolated feedback, the complete workflow is examined using data, user signals and **Quality Engineering** methods.

03 USER BEHAVIOUR, WORKFLOW DATA AND FEEDBACK

The first stage combines three sources of quality information: user behaviour, workflow data and direct user feedback. Behavioural signals can reveal where users experience friction, repeat actions or spend additional time. Workflow data can help identify stages associated with delay, errors or rework, while direct feedback provides context behind those observations.

These signals are analysed together rather than treating one source as the complete representation of quality. Once a significant quality concern is identified, the improvement process enters a structured **PDCA** cycle to support systematic planning, experimentation, evaluation and refinement.

04 ROOT CAUSE ANALYSIS

After identifying a major quality concern, **Pareto Analysis** can be used to determine which workflow problems contribute most significantly to the overall issue. This prevents improvement efforts from being spread across numerous minor problems when a smaller number of dominant contributors may have a greater impact.

The major contributor can then be investigated using **5 Why** and **Fishbone Analysis** to explore possible causes across relevant categories. **Shainin Red X** can subsequently support identification of the dominant cause responsible for the observed variation or performance problem. The objective is to move from symptom identification toward evidence based root cause understanding.

05 PDCA AND QUALITY IMPROVEMENT

Once the dominant cause is understood, the improvement enters a structured **PDCA** cycle. The Plan stage defines the problem, relevant quality measures and improvement direction. The Do stage introduces the proposed change through controlled experimentation. The Check stage evaluates whether the change produces the expected improvement without creating unacceptable quality problems, while the Act stage focuses on refining and standardising a validated solution.

This creates a disciplined connection between analysis and implementation, ensuring that improvement decisions are supported by evidence rather than assumptions.

06 TAGUCHI ROBUST DESIGN AND DOE

A solution that performs well under one condition does not automatically guarantee consistent performance across different users or operating conditions. Taguchi's Robust Design approach is therefore adapted here as a **Quality Engineering** perspective for developing a solution that remains stable despite variation.

Design of Experiments can be used to systematically evaluate important factors and their combinations rather than relying on isolated trials. The purpose is not simply to find a solution that produces a good result once, but to identify a more robust solution that can maintain acceptable performance across relevant conditions.

07 FMEA BASED RISK EVALUATION

Before implementation, the proposed improvement is evaluated through **FMEA**. Potential failure modes are considered using Severity, Occurrence and Detection to establish an **RPN** based risk perspective.

The purpose of **FMEA** is to identify possible problems before they reach users and to determine where preventive controls may be required. This creates a quality gate between experimentation and deployment, allowing risks associated with the proposed solution to be considered before wider implementation.

08 CONTROLLED VALIDATION ACROSS USERS

After experimentation and risk evaluation, the proposed solution should not immediately move to broad deployment. Validation should include different user types and different usage conditions so that the solution can be evaluated beyond average performance.

Controlled validation helps identify variation that may remain hidden during a limited experiment and provides additional evidence before wider deployment. The structure moves from small controlled validation to broader validation and controlled wider rollout.

THREE STAGE VALIDATION STRUCTURE

STAGE 1: SMALL CONTROLLED VALIDATION → STAGE 2: BROADER VALIDATION → STAGE 3: CONTROLLED WIDER ROLLOUT

09 CONTINUOUS QUALITY MONITORING

After rollout, quality should continue to be monitored through user feedback, behavioural signals and relevant workflow measures. The objective is to identify whether a previously controlled issue begins to return or whether a new quality concern emerges.

If a new problem appears, the process returns to root cause analysis and the **PDCA** cycle. The improvement is then re evaluated before being standardised again, creating a quality system capable of learning from changing conditions.

10 KAIZEN AND STANDARDISATION

Quality should not be treated as a one time achievement. Continuous improvement requires regular feedback, small meaningful changes, validation of those changes and standardisation of improvements that demonstrate consistent results.

This reflects the Japanese **Kaizen** philosophy, where quality develops through continuous incremental improvement. A validated improvement becomes part of the standard process while new signals trigger further analysis.

CONTINUOUS IMPROVEMENT VISUAL

FEEDBACK → ANALYSIS → IMPROVEMENT → VALIDATION → STANDARDISATION → MONITORING

11 QUALITY ENGINEERING AND DIGITAL PRODUCT QUALITY

This framework connects user centred product thinking with **Quality Engineering** methodology. User behaviour and feedback provide quality signals, analytical tools identify major contributors, Taguchi Robust Design supports experimentation, **FMEA** evaluates risks, **PDCA** structures improvement and controlled validation verifies performance across different conditions.

"Quality in a digital creative workflow is not achieved by improving one feature alone. It is achieved by understanding user behaviour, identifying dominant causes, designing robust solutions, evaluating implementation risks, validating across variation and continuously improving through evidence."

Disclaimer: This case study is an independent professional research exercise by Jagriti Singh. It presents a conceptual application of **Quality Engineering** methodologies to a digital creative workflow environment and is not an official Adobe project, internal Adobe analysis, audit, specification, endorsement, partnership or affiliation.